

Student Management System - CRUD Application

A full-stack Single Page Application (SPA) built with React, Firebase Firestore, and React Router DOM.

Project Overview

This is a comprehensive CRUD (Create, Read, Update, Delete) application for managing student information. The application demonstrates:

- **SPA Routing:** Using React Router DOM for seamless navigation
- **Firebase Firestore:** Cloud database for data persistence
- **Dynamic Routing:** View and edit individual student records
- **Real-time UI Updates:** Immediate feedback after database operations

Features

- ✚ **Create:** Add new student records with form validation
- 📄 **Read:** View all students in a card-based layout
- 👤 **Read Single:** Dynamic routing to view individual student details
- ⇒ **Update:** Edit existing student records with pre-filled forms
- 🗑️ **Delete:** Remove student records with confirmation

Tech Stack

- **Frontend:** React 19.2.4, React Router DOM 7.15.0
- **Backend:** Firebase Firestore
- **Build Tool:** Vite 8.0.1
- **Styling:** CSS3
- **Deployment:** Firebase Hosting

Installation

```
# Clone the repository
git clone <your-repo-url>
cd my-app

# Install dependencies
npm install

# Set up Firebase (already configured)
# Update src/firebase/firebaseConfig.js if needed with your Firebase credentials
```

Running the Application

Development Mode

```
npm run dev
```

The app will be available at <http://localhost:5173>

Production Build

```
npm run build
npm run preview
```

Deployment

```
firebase deploy
```

Project Structure

```
my-app/
├── src/
│   ├── components/
│   │   ├── layouts/
│   │   │   └── Navbar.jsx
│   │   ├── pages/
│   │   │   ├── HomePage.jsx
│   │   │   ├── CreateItemPage.jsx
│   │   │   ├── ViewAllPage.jsx
│   │   │   ├── ViewSinglePage.jsx
│   │   │   ├── EditItemPage.jsx
│   │   │   ├── PageNotFound.jsx
│   │   └── StudentForm.jsx
│   ├── firebase/
│   │   └── firebaseConfig.js
│   ├── App.jsx
│   ├── main.jsx
│   └── App.css
├── public/
├── dist/
├── firebase.json
├── vite.config.js
└── package.json
```

Routes

Route	Component	Purpose
/	HomePage	Landing page with action buttons
/create	CreateItemPage	Form to add new students
/items	ViewAllPage	Display all students
/items/:id	ViewSinglePage	View single student details
/edit/:id	EditItemPage	Edit student record
*	PageNotFound	404 error page

Firestore Collection

Collection: students

Document Structure:

```
{
  "name": "Student Name",
  "rollNo": "Roll Number",
  "department": "BSCS",
  "semester": "4",
  "createdAt": "timestamp"
}
```

Environment Setup

The Firebase configuration is located in `src/firebase/firebaseConfig.js`:

- Project ID: `pixel-eee3d`
- Authentication: Firebase Rules configured
- Hosting: Firebase Hosting enabled

Features Implemented

- ☒ Task 1: SPA Routing with React Router DOM
 - Multiple routes with dynamic parameters
 - Consistent Navbar layout
 - No page reloads during navigation
- ☒ Task 2: CRUD Operations with Firestore
 - Create: Form submission to Firestore
 - Read: Fetch and display all documents
 - Read (Single): Dynamic routing with document fetching
 - Update: Pre-filled form with Firestore update
 - Delete: With UI confirmation and immediate update
- ☒ Task 3: Deployment
 - Deployed on Firebase Hosting
 - Live URL: `https://pixel-app-b4761.web.app`
- ☒ Task 4: GitHub Repository
 - Complete source code with proper structure
 - Comprehensive README documentation

Live URL

 Hosted Application: `https://pixel-app-b4761.web.app`

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Author

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